



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

**SECP3623
Database Programming
2025/2026 SEMESTER 1
Lab Assignment 2**

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Academic Integrity

All students must work INDIVIDUALLY. Use of AI-generated SQL or code sharing is strictly prohibited. Identical or AI-generated scripts will receive **zero marks**.

Provide screenshots of every instruction in all tasks.

SUBMISSION REQUIREMENTS

Submit one (1) ZIP file containing:

1. Lab2_Clinic_Matric Number.txt (Copy from MongoDB Shell - Mongosh)
 - All MongoDB commands used
 - Clearly labelled by question number
2. Lab2_Clinic_Matric Number.pdf
 - Screenshots of command outputs
 - Each screenshot must clearly show the command and result

PART A - DATA PREPARATION (3 MARKS)

Question A1 (1 mark)

```
> use clinicdb
< switched to db clinicdb
> db.createCollection("appointments")
< { ok: 1 }
> db.createCollection("loginSessions")
< { ok: 1 }
```

Question A2 (2 marks)

- appointments collection

```
> db.appointments.insertMany([
  {
    appointmentId: "A001",
    patientName: "Ahmad",
    doctor: "Dr. Ali",
    department: "Cardiology",
    status: "COMPLETED",
    fee: 250,
    appointmentDate: ISODate("2025-05-01")
  },
  {
    appointmentId: "A002",
    patientName: "Siti",
    doctor: "Dr. Lee",
    department: "Dermatology",
    status: "PENDING",
    fee: 120,
    appointmentDate: ISODate("2025-05-03")
  },
  {
    appointmentId: "A003",
    patientName: "Abu",
    doctor: "Dr. Karim",
    department: "Orthopedics",
    status: "COMPLETED",
    fee: 300,
    appointmentDate: ISODate("2025-05-04")
  },
])
```

```

{
  appointmentId: "A004",
  patientName: "Nurul",
  doctor: "Dr. Farah",
  department: "Cardiology",
  status: "CANCELLED",
  fee: 180,
  appointmentDate: ISODate("2025-05-05")
},
{
  appointmentId: "A005",
  patientName: "Aina",
  doctor: "Dr. Lee",
  department: "Dermatology",
  status: "COMPLETED",
  fee: 100,
  appointmentDate: ISODate("2025-05-06")
},
{
  appointmentId: "A006",
  patientName: "Hafiz",
  doctor: "Dr. Karim",
  department: "Orthopedics",
  status: "COMPLETED",
  fee: 150,
  appointmentDate: ISODate("2025-05-07")
}
])

```

```

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696e04c0fbb1cf4516c6db1a'),
    '1': ObjectId('696e04c0fbb1cf4516c6db1b'),
    '2': ObjectId('696e04c0fbb1cf4516c6db1c'),
    '3': ObjectId('696e04c0fbb1cf4516c6db1d'),
    '4': ObjectId('696e04c0fbb1cf4516c6db1e'),
    '5': ObjectId('696e04c0fbb1cf4516c6db1f')
  }
}

```

- loginSessions collection

```
> db.loginSessions.insertMany([
  {
    staffId: "S001",
    staffName: "Aina",
    role: "Nurse",
    status: "ACTIVE",
    ipAddress: "192.168.1.10",
    loginTime: ISODate("2025-05-07T08:30:00")
  },
  {
    staffId: "S002",
    staffName: "Hakim",
    role: "Doctor",
    status: "ACTIVE",
    ipAddress: "192.168.1.11",
    loginTime: ISODate("2025-05-07T09:00:00")
  },
  {
    staffId: "S003",
    staffName: "Sofia",
    role: "Receptionist",
    status: "INACTIVE",
    ipAddress: "192.168.1.12",
    loginTime: ISODate("2025-05-06T17:00:00")
  },
])
```

```
{
  staffId: "S004",
  staffName: "Adam",
  role: "Admin",
  status: "ACTIVE",
  ipAddress: "192.168.1.13",
  loginTime: ISODate("2025-05-07T10:15:00")
},
{
  staffId: "S005",
  staffName: "Nabila",
  role: "Doctor",
  status: "INACTIVE",
  ipAddress: "192.168.1.11",
  loginTime: ISODate("2025-05-05T16:45:00")
},
{
  staffId: "S006",
  staffName: "Irfan",
  role: "Nurse",
  status: "ACTIVE",
  ipAddress: "192.168.1.14",
  loginTime: ISODate("2025-05-07T11:00:00")
}
])
```

```
    acknowledged: true,
    insertedIds: {
      '0': ObjectId('696e0584fbb1cf4516c6db20'),
      '1': ObjectId('696e0584fbb1cf4516c6db21'),
      '2': ObjectId('696e0584fbb1cf4516c6db22'),
      '3': ObjectId('696e0584fbb1cf4516c6db23'),
      '4': ObjectId('696e0584fbb1cf4516c6db24'),
      '5': ObjectId('696e0584fbb1cf4516c6db25')
    }
  }
}
```

PART B - APPOINTMENT REPORTING (7 MARKS)

Question B1 - Advanced Query s Sorting (3 marks)

```
> db.appointments.find(
  {
    status: "COMPLETED",
    fee: { $gte: 100, $lte: 300 }
  },
  {
    _id: 0,
    appointmentId: 1,
    patientName: 1,
    department: 1,
    fee: 1
  }
).sort(
  {
    department: 1,
    fee: -1
  }
)
```

```

< {
  appointmentId: 'A001',
  patientName: 'Ahmad',
  department: 'Cardiology',
  fee: 250
}
{
  appointmentId: 'A005',
  patientName: 'Aina',
  department: 'Dermatology',
  fee: 100
}
{
  appointmentId: 'A003',
  patientName: 'Abu',
  department: 'Orthopedics',
  fee: 300
}
{
  appointmentId: 'A006',
  patientName: 'Hafiz',
  department: 'Orthopedics',
  fee: 150
}

```

Question B2 - Aggregation Summary (4 marks)

```

> db.appointments.aggregate([
  {
    $group: {
      _id: "$department",
      totalAppointments: { $sum: 1 },
      totalFees: { $sum: "$fee" }
    }
  },
  {
    $sort: { totalFees: -1 }
  }
])

```

```
< {
  _id: 'Orthopedics',
  totalAppointments: 2,
  totalFees: 450
}
{
  _id: 'Cardiology',
  totalAppointments: 2,
  totalFees: 430
}
{
  _id: 'Dermatology',
  totalAppointments: 2,
  totalFees: 220
}
```

PART C - CLINIC SECURITY MONITORING (6 MARKS)

Question C1 - Active Staff Sessions (3 marks)

```
> db.loginSessions.find(
  { status: "ACTIVE" },
  {
    _id: 0,
    staffName: 1,
    role: 1,
    ipAddress: 1,
    loginTime: 1
  }
).sort(
  { loginTime: -1 }
)
```

```

< {
  staffName: 'Irfan',
  role: 'Nurse',
  ipAddress: '192.168.1.14',
  loginTime: 2025-05-07T11:00:00.000Z
}
{
  staffName: 'Adam',
  role: 'Admin',
  ipAddress: '192.168.1.13',
  loginTime: 2025-05-07T10:15:00.000Z
}
{
  staffName: 'Hakim',
  role: 'Doctor',
  ipAddress: '192.168.1.11',
  loginTime: 2025-05-07T09:00:00.000Z
}
{
  staffName: 'Aina',
  role: 'Nurse',
  ipAddress: '192.168.1.10',
  loginTime: 2025-05-07T08:30:00.000Z
}

```

Question C2 - Distinct s Count (3 marks)

a) List all distinct IP addresses used by staff with ACTIVE sessions. (1.5 marks)

```

> db.loginSessions.distinct(
  "ipAddress",
  { status: "ACTIVE" }
)
< [ '192.168.1.10', '192.168.1.11', '192.168.1.13', '192.168.1.14' ]

```


b) Calculate the number of unique staff members who currently have ACTIVE sessions. (1.5 marks)

```
> db.loginSessions.distinct(
  "staffId",
  { status: "ACTIVE" }
).length
< 4
```

PART D - INDEXING s DATA MAINTENANCE (4 MARKS)

Question D1 - Compound Index (2 marks)

```
> db.appointments.createIndex(
  { department: 1, status: 1 }
)
< department_1_status_1
> db.appointments.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { department: 1, status: 1 },
    name: 'department_1_status_1'
  }
]
```

Question D2 - Update s Delete Operations (2 marks)

a) Update all appointments with status CANCELLED to ARCHIVED. (1 mark)

```
> db.appointments.updateMany(
  { status: "CANCELLED" },
  { $set: { status: "ARCHIVED" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

b) Delete all login session records with status INACTIVE. (1 mark)

```
> db.loginSessions.deleteMany(  
  { status: "INACTIVE" }  
)  
< {  
  acknowledged: true,  
  deletedCount: 2  
}
```

MARKING RUBRICS

Part	Question	Assessment Criteria	Marks	
A	A1	Database clinicdb created correctly	0.5	
		Both collections (appointments, loginSessions) created	0.5	
	A2	Minimum 6 valid documents inserted into appointments	1.0	
		Minimum 6 valid documents inserted into loginSessions	1.0	
		Part A Subtotal	3.0	
B	B1	Correct filtering conditions (status COMPLETED, fee range 100–300)	1.5	
		Correct projection of required fields	0.5	
		Correct sorting (department ASC, fee DESC)	1.0	
	B2	Correct aggregation grouping by department	1.5	
		Correct aggregation calculations (count C total fees)	1.5	
		Correct sorting by total fees (DESC)	1.0	
		Part B Subtotal	7.0	
C	C1	Correct filtering of ACTIVE sessions	1.0	
		Correct projection of required fields	1.0	
		Correct sorting by latest login time	1.0	
	C2(a)	Correct retrieval of distinct ACTIVE IP addresses	1.5	
	C2(b)	Correct count of unique ACTIVE staff	1.5	
		Part C Subtotal	6.0	
D	D1	Correct compound index fields (department, status)	1.5	
		Index created successfully and shown	0.5	
	D2(a)	Correct update operation (CANCELLED → ARCHIVED)	1.0	
	D2(b)	Correct delete operation (INACTIVE sessions)	1.0	
		Part D Subtotal	4.0	
		TOTAL	20 MARKS	